Customer Shopping Behavior Analysis

A comprehensive data-driven study of 3,900 transactional records to decode consumer patterns and strategic growth opportunities.

EXECUTIVE SUMMARY

Analysis Objectives

Uncover spending patterns, product preferences, and demographic behaviors to drive strategic business decisions.

Identify customer segments (Loyal, New, Returning) to optimize marketing ROI and retention programs.

Key Business Impact

Evaluation of subscription dynamics and the correlation between frequency of purchase and membership status.

Optimization of discount strategies based on high-spending behaviors and review ratings.

DATASET OVERVIEW

Structure & Scope

Record Count: 3,900 purchase entries.

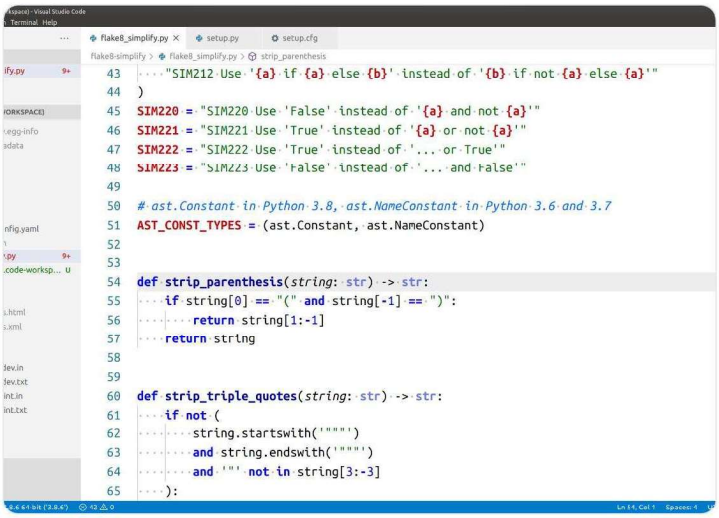
Dimensions: 18 key behavioral and demographic columns.

Quality Check: Review ratings missing 37 values (handled via category-median imputation).

METHODOLOGY

Technical Workflow

- **Cleaning (Python):** Pandas-based normalization, snake_case conversion, and missing value handling.
- **DB Integration:** Cleaned data exported to PostgreSQL for advanced relational querying.
- **Feature Engineering:** Binning customers by age groups and calculating purchase frequency days.



GENDER REVENUE MIX

Economic Contribution

Male Segment: Leads with $157,890 in total revenue generated.

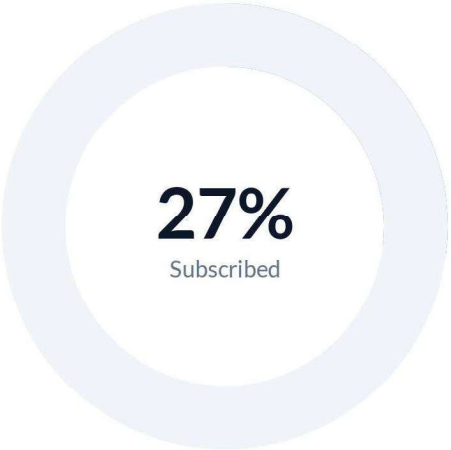
Female Segment: Contributed $75,191 recorded revenue.

Analysis identifies a significant opportunity to scale engagement within the female demographic through targeted assortments.

REVENUE BY AGE COHORT

Young Adults are the primary revenue driver, accounting for over $62k in sales.

SUBSCRIPTION STATUS



Yes (1,053 Users)

No (2,847 Users)

Avg Spend: ~$59.50

INVENTORY & SENTIMENT

Top Volume Leaders

01	Jewelry / Blouse	171 Orders
02	Pants	171 Orders
03	Shirt	169 Orders
04	Dress	166 Orders

Core clothing staples drive the highest transactional volume, indicating consistent demand for everyday apparel.

CUSTOMER SATISFACTION

3.86 / 5.0

Gloves: Highest rated item, reflecting superior quality perception.

PROMO SENSITIVITY

50%

Hats: Half of all sales involve discounts, highlighting high price sensitivity.

CATALOG HEALTH

3.75

Global average review rating remains stable across 25 product items.

LOGISTICS PERFORMANCE

Fulfillment Impact

Express shipping users exhibit a higher willingness to spend, with an average purchase value of **$60.48**.

Standard shipping follows at a **$58.46** average spend.

Strategy: Leverage premium shipping speed as a value-add for high-spending customer segments.

CATEGORY PRODUCT LEADERS

Category	Rank #1 Product	Rank #2 Product	Total Orders
Accessories	Jewelry	Sunglasses / Belt	171 / 161
Clothing	Blouse / Pants	Shirt	171 / 169
Footwear	Sandals	Shoes / Sneakers	160 / 150
Outerwear	Jacket	Coat	163 / 161

STRATEGIC ROADMAP

Boost Subscriptions

Launch targeted benefits for the 73% non-subscriber base to increase LTV.

Loyalty Rewards

Transition 701 Returning buyers into the Loyal segment via exclusive tiers.

Targeted Marketing

Focus campaigns on Young Adult cohorts and Express shipping enthusiasts.

Questions?

Thank you for your review. This conclude the analysis of the Customer Shopping Behavior dataset.